SATYA ROHAN KOMPELLA

Software Engineering Intern | Computer Vision & AI/ML

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EXPERIENCE

Project Intern — JNTU Hyderabad 05/2026 – Present

Hyderabad, Telangana

- Engineered a core module of an 18-month DST-NGP-funded 3D urban infrastructure inspection system, applying computer vision, photogrammetry, and 3D visualization techniques.
- Processed and analysed a multi-modal dataset of 600 LiDAR scans and 2,000+ RGB images captured from the Pune Bridge, enabling large-scale structural inspection.
- Built end-to-end geospatial reconstruction and anomaly detection pipelines using Pix4Dmatic, QGIS, Blender, and YOLOv8 to automate structural defect localization.
- Trained and fine-tuned YOLOv8 object detection (mAP@50 = 0.77) and segmentation models on large-scale datasets, improving structural anomaly detection accuracy.

SELECTED PROJECTS

ROST — Secure Linux Examination Environment 05/2026 – Present

- Architected a custom Ubuntu-based examination operating system with a locked-down secure exam mode.
- Automated application restrictions and system configuration via Bash scripts, streamlining secure deployment for educational institutions.
- Customized the Linux desktop environment to support scalable, secure educational deployments.

VIRA — AI Virtual Assistant 05/2025 – 08/2025

- Built a Python-based AI virtual assistant supporting automation, web search, weather, news, and voice interaction.
- Designed a multi-layer reasoning pipeline to handle logical analysis and complex, multi-step queries.
- Implemented real-time voice input/output for natural, human-like interaction.
- Deployed the application on Hugging Face, growing adoption to 80+ active users within the first month.

Deep-Fake Image Analyzer 12/2025 – 01/2026

- Developed a real-time deepfake detection system by fine-tuning an XceptionNet model on the FaceForensics++ dataset.
- Built a scalable FastAPI backend to handle model inference and API requests efficiently.
- Developed a React frontend enabling real-time image analysis and prediction visualization.

SKILLS

Programming Languages: Python, Java, C

AI/ML: Generative AI APIs, YOLOv8, PyTorch, OpenCV

Web Development: React.js, FastAPI, SQLAlchemy, HTML, CSS, JavaScript, Firebase

Databases: PostgreSQL, Oracle SQL

Tools: Git, GitHub, Tableau, Linux, Bash, Pix4Dmatic, QGIS, Blender

Core CS: Data Structures & Algorithms, OOP, DBMS, OS, Computer Networks, Compiler Design

PUBLICATIONS

"UAV-Based 3D Bridge Reconstruction and Geospatial Anomaly Localization for Structural Health Monitoring" -**Accepted Abstract (ID: ACRS26_ABS_M517429), ACRS 2026 (Asian Conference on Remote Sensing).**

EDUCATION

B.Tech in Computer Science & Engineering — Raghu Engineering College (Autonomous) 07/2024 – 06/2028

Visakhapatnam, Andhra Pradesh, India *CGPA: 8.92*

CERTIFICATIONS

- AWS AI/ML Scholars -Udacity
- AWS Solutions Architecture -Forage
- Tata Data Analytics -Forage